

期中复习



Outline

- Linear, Circular, Higher dimensional motions
- Newton's Laws
- Energy, Momentum, Angular momentum, Conservation Laws
- Rigid body rotation
- Gravitation
- Oscillation
- Wave

One big difference compared to high-school physics: Advanced mathematical tools

- Limit: $\Delta t \rightarrow 0, \Delta x \rightarrow 0, \dots$

- Differentiation: $v = \dot{x} = \frac{dx}{dt}$
 $a = \ddot{x} = \frac{dv}{dt} = \frac{d^2x}{dt^2}, \dots$

- Integration: $\Delta U = - \int_{x_i}^{x_f} F(x) dx$
 $I = \int r^2 dm, \dots$

- Vectors: $\vec{r}, \vec{v}, \vec{a}$

Vector decomposition: $\vec{a} = \sum_i a_i \hat{e}_i = \{\hat{e}_1, \hat{e}_2, \hat{e}_3\} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$

Vector addition: $\vec{a} + \vec{b}$

Scalar product: $\vec{a} \cdot \vec{b}$

Cross product: $\vec{a} \times \vec{b}$

Unit Vector: $\hat{a} = \vec{a}/|\vec{a}|$

Coordinates

- Cartesian: $\hat{i}, \hat{j}, \hat{k}$ or $\hat{e}_x, \hat{e}_y, \hat{e}_z$
- Polar: $\hat{r}, \hat{\phi}$ ($r dr d\phi$)
- Spherical: $\hat{r}, \hat{\theta}, \hat{\varphi}$ ($r^2 \sin \theta dr d\theta d\varphi$)
- Orthonormalized or not...

Significant figures

- Significant figures

Addition/Subtraction

Multiplication/Division

- Uncertainty propagation

$$\Delta y = \sqrt{\left(\frac{\partial y}{\partial x_1}\right)^2 (\Delta x_1)^2 + \cdots + \left(\frac{\partial y}{\partial x_n}\right)^2 (\Delta x_n)^2}$$

Motions

- Position: $\vec{r}$
- Velocity: $\vec{v} = \frac{d\vec{r}}{dt}$
- Acceleration: $\vec{a} = \frac{d\vec{v}}{dt}$

$$\vec{r} = \sum_i r_i \hat{e}_i$$

$$\frac{d\vec{r}}{dt} = \sum_i \left(\frac{dr_i}{dt} \hat{e}_i + r_i \frac{d\hat{e}_i}{dt} \right)$$

Newton's Laws

- First Law: no force, no velocity change (inertia)

- Second Law: $\vec{F} = m\vec{a}$

- Third Law: $\vec{F}_{12} = -\vec{F}_{21}$

Free body diagram analysis

Inertial and non-inertial frames

- Inertial frame: where Newton's Laws hold

Galilean transformation: (from inertial frame to inertial frame)

$$\vec{r}' = \vec{r} - \vec{v}_0 t$$

$$\vec{v}' = \vec{v} - \vec{v}_0$$

$$\vec{a}' = \vec{a}$$

- Non-inertial frame:

Relative to an accelerating car, moving along a circle, ...

Fictitious force $\vec{F}_{\text{fictitious}} = -m\vec{a}$ (if no relative motion)

Work, Kinetic Energy

Work: $W = \vec{F} \cdot \vec{d} = Fd \cos \theta$ (constant force)

$$W = \int_{\vec{r}_i}^{\vec{r}_f} \vec{F}(\vec{r}) \cdot d\vec{r} \quad (\text{varying force})$$

Work – Kinetic energy theorem:

$$W = K_f - K_i$$

Kinetic energy:

$$K = \frac{1}{2}mv^2$$

Mechanical Energy Conservation

Mechanical energy conserved if only conservative forces are present:

$$K + U = \text{constant}$$

- Conservative force: work done around any closed path is zero.

e.g., $F = -kx$, $F = mg$

- Nonconservative force.

e.g., $F = \mu_k F_N$

Total Energy conservation in presence of nonconservative force?

Potential energy

$$U_f - U_i = -W$$

$$= - \int_{\vec{r}_i}^{\vec{r}_f} \vec{F}(\vec{r}) \cdot d\vec{r}$$

Well-defined only for conservative forces

e.g., Gravitation near surface of earth:

$$U = mgy$$

$$F = -\frac{dU}{dy} = -mg$$

e.g., Gravitation in general:

$$U = -\frac{Gm_1m_2}{r}$$

$$F = -\frac{dU}{dr} = -\frac{Gm_1m_2}{r^2}$$

Impulse, Momentum

Impulse: $\vec{I} = \int_{t_i}^{t_f} \vec{F} dt$

Impulse – Momentum theorem:

$$\vec{I} = \vec{p}_f - \vec{p}_i$$

(Linear) Momentum:

$$\vec{p} = m\vec{v}$$

Momentum Conservation

(Linear) Momentum conserved if the system is

- Closed (no particles leave or enter)
- Isolated (no external force)

$$\sum_i \vec{p}_i = \text{constant}$$

e.g., collisions (elastic, inelastic):

- Total momentum conserved
- Total mechanical energy depends (elastic: yes; inelastic: no)

Gravitational Laws

Gravitational force:

$$F = \frac{Gm_1m_2}{r^2}$$

Kepler's laws:

- Elliptical orbits
- Equal area during equal time interval
- Orbital period $T^2 \propto a^3$

Total energy in satellite orbit:

$$E = -\frac{GMm}{2r}$$

Circular Motion, Rotation

Rotation axis



$$\left\{ \begin{array}{l} \theta \\ \omega = \frac{d\theta}{dt} \\ \alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} \end{array} \right.$$

$$\vec{v} = \vec{\omega} \times \vec{r} \quad (\text{right-hand rule})$$

$$a_t = \alpha r$$

$$a_r = \omega^2 r = \frac{v^2}{r}$$

Pure rotation: $s = R\theta$

Torque, Moment of Inertia

$$\vec{\tau} = \vec{r} \times \vec{F} \quad (\text{right-hand rule})$$

$$\vec{\tau} = I\alpha\hat{\omega} \quad (\text{analog of } \vec{F} = m\vec{a})$$

$$\tau = I\alpha$$

Moment of inertia:

$$I = \sum_i m_i r_i^2 \quad (\text{discrete})$$

$$= \int (dm) r^2 \quad (\text{continuous})$$

Rotation vs Linear Motion

$$\omega = d\theta/dt$$

$$v = dx/dt$$

$$\alpha = d\omega/dt$$

$$a = dv/dt$$

$$\tau = I\alpha$$

$$F = ma$$

$$W = \int_{\theta_i}^{\theta_f} \tau d\theta$$

$$W = \int_{x_i}^{x_f} F dx$$

$$K_R = \frac{1}{2} I \omega^2$$

$$K = \frac{1}{2} m v^2$$

Angular Momentum

Angular Momentum: $\vec{L} = \vec{r} \times \vec{p}$ (remember $\vec{\tau} = \vec{r} \times \vec{F}$)

$$\vec{L} = I\vec{\omega} \quad (\text{moment of inertia tensor})$$

Analog of impulse -- linear momentum theorem:

$$\vec{\tau}_{\text{ext}} dt = d\vec{L}$$

Angular momentum is conserved if total external torque is zero:

$$\vec{L} = \text{constant}$$

Center of Mass

$$\vec{r}_{\text{CM}} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i} \qquad \sum_i \vec{F}_i = M \vec{a}_{\text{CM}}$$

König theorem: $K = K_{\text{CM}} + K'$

$$= \frac{1}{2} M v_{\text{CM}}^2 + \frac{1}{2} \sum_i m_i (\vec{v}_i - \vec{v}_{\text{CM}})^2$$

Total momentum in CM frame: $\sum_i m_i (\vec{v}_i - \vec{v}_{\text{CM}}) = 0$

Relative motion: $\sum_i \vec{r}'_i \times \vec{F}_{i,\text{ext}} = \frac{d}{dt} \left(\sum_i \vec{r}'_i \times \vec{p}'_i \right)$

Parallel axis theorem: $I = I_{\text{CM}} + M h^2$

Harmonic Motion

Stable equilibrium:

$$F = -\frac{dU}{dx} = 0$$
$$\frac{d^2U}{dx^2} > 0$$

Potential energy around a stable equilibrium point:

$$U(x) = U(x_0) + \frac{1}{2} \left. \frac{d^2U}{dx^2} \right|_{x_0} (x - x_0)^2 + \dots$$
$$\approx U(x_0) + \frac{1}{2} k (x - x_0)^2$$

Simple harmonic motion:

$$x - x_0 = A \cos(\omega t + \phi)$$

$$\omega = \sqrt{\frac{k}{m}}$$

Motion of multiple objects

- Modes counting

- Eigen modes

$$x_i = A_i \cos(\omega t + \phi)$$

(eigen values and eigen vectors indicate the basic oscillatory motions)

- Overall motion: superposition of eigen modes

Waves

- Transverse
- Longitudinal (e.g., sound)

Right moving: $f(x - vt)$

Left moving: $f(x + vt)$

Principle of superposition (linear waves):

$$y'(x, t) = y_1(x, t) + y_2(x, t)$$

- Reflection: v opposite sign
- Transmission: v same sign

Sinusoidal Waves

Linear wave equation:

$$\frac{\partial^2 u}{\partial t^2} = v^2 \frac{\partial^2 u}{\partial x^2}$$

Sinusoidal waves:

$$y = A \sin(kx - \omega t + \phi)$$

$$v = \frac{\omega}{k} = \frac{\lambda}{T} \quad (\text{wave speed})$$

Note:

$$v \neq v_y$$

$$v_y \equiv \frac{\partial y}{\partial t}$$

Sinusoidal Waves

Interference:

- Constructive $\Delta\phi = 2\pi n$
- Destructive $\Delta\phi = 2\pi(n + 1/2)$

Different frequencies: Beat

$$\lambda' \sim \frac{2\pi v}{\Delta\omega/2} \quad (\text{modulating wavelength})$$

$$f_{\text{beat}} = \frac{\Delta\omega}{2\pi} \quad (\text{beat frequency})$$

Standing Waves

$$y_1 = A \sin(kx - \omega t)$$

$$y_2 = A \sin(kx + \omega t)$$

$$y = y_1 + y_2 = 2A \sin(kx) \cos(\omega t)$$

- Nodes: $kx = n\pi$
- Antinodes: $(\pi/2 \text{ from nodes})$

Standing wave with both ends fixed:

$$\lambda_n = \frac{2L}{n}$$

$$f_n = \frac{v}{\lambda_n} = n \frac{v}{2L}$$

Sound Waves

- Wave speed: $v = \sqrt{\frac{B}{\rho}}$
- Intensity: $I = \frac{P}{A} = \frac{1}{2} \rho v \omega^2 s_m^2$
- Sound level: $\beta = 10 \lg (I / I_0)$
- Sound frequency: infrasound, audible, ultrasound
- Doppler effect: $f' = f \frac{v \pm v_D}{v \pm v_S}$

Relax

- Nothing beyond lecture notes (handwritten) + homework
- Midterm + Quiz: 30%